Harmonic Balance Domain Decomposition Finite Elements for Nonlinear Passive Microwave Devices Analysis

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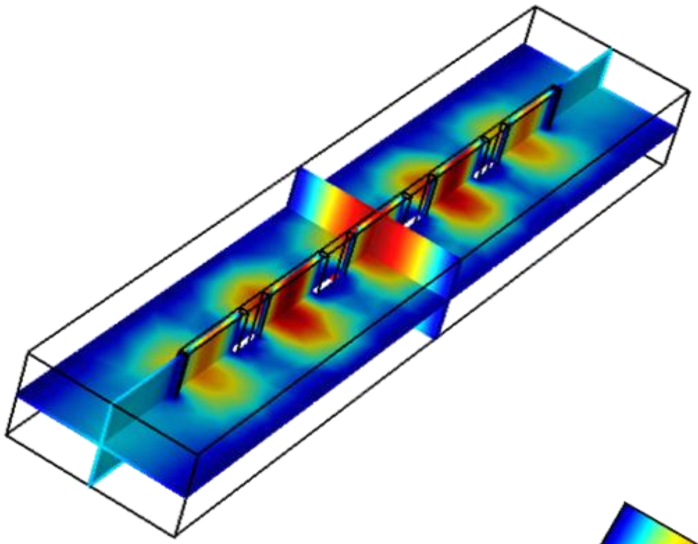
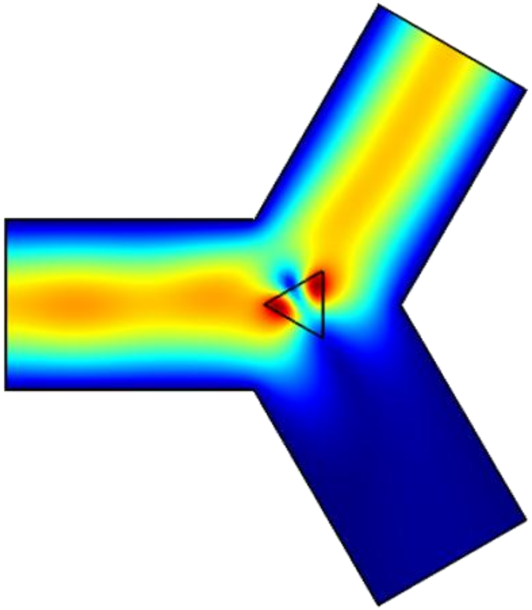
- Motivation
- Formulation
  - Harmonic Balance
  - Domain Decomposition
- Numerical experiments
  - Nonlinear permittivity
  - Nonlinear permeability
- Conclusion





- When high power densities are involved within microwave devices, materials exhibit a significant nonlinear behavior:

$$\mathbf{D} = [\bar{\bar{\varepsilon}}(\mathbf{E},\mathbf{H})]\mathbf{E}$$
$$\mathbf{B} = [\bar{\bar{\mu}}(\mathbf{E},\mathbf{H})]\mathbf{H}$$

- If steady state fields are aimed at for a finite set of frequencies, a frequency domain analysis might be preferable to a time domain analysis

- Frequency domain methods for solving nonlinear PDEs require an iterative solution and the inclusion of higher order harmonics to improve the accuracy

- Depending on the field intensities and the nonlinear operator that describes the material behavior, the nonlinear solving loop might be computationally expensive

Harmonic Balance

- A multiharmonic finite elements formulation, the Harmonic Balance Finite Elements (HBFE), allows for the inclusion of higher order harmonics which are coupled by means of material characteristics
- The HBFE approach also allows for passive intermodulation analysis (3^{rd} order intermodulation products)

Domain Decomposition

- For designs that include small parts of materials that present significant nonlinear behavior at high intensity fields, domain decompositions techniques might be used to restrict the nonlinear solving loop to fewer DoFs

Conventional FE (single harmonic)

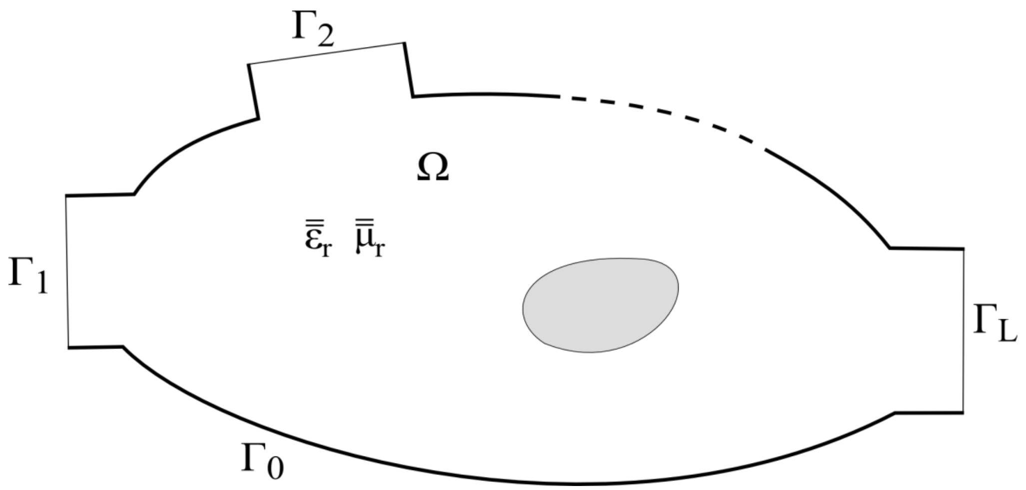
We wish to solve the vector Helmholtz equation

$$\nabla\times\left(\overline{\overline{\mu}}_r^{-1}(\mathbf{E}(\mathbf{r}),\mathbf{r})\nabla\times\mathbf{E}(\mathbf{r})\right)-k_0^2\overline{\overline{\varepsilon}}_r(\mathbf{E}(\mathbf{r}),\mathbf{r})\mathbf{E}(\mathbf{r})=0 \qquad \text{in } \Omega$$

with perfect electric conditions on Γ_0 and modal port expansions on Γ_l, l=1…L.

The electric field is expanded in terms of basis functions and a Galerkin projection (weak form) leads to a linear system of equations

$$[a_{mn}][E_n]=[b_m]$$



Harmonic Balance FE (multiple harmonics)

- The electric field is expressed as a sum of harmonic functions:

$$\hat{\mathbf{E}}(\mathbf{r},t) = \sum_{p=1}^{P}\left(\mathbf{E}_p^{(s)}(\mathbf{r})\sin(\omega_p t) + \mathbf{E}_p^{(c)}(\mathbf{r})\cos(\omega_p t)\right)$$

- Materials permittivity or permeability is also expressed in terms of harmonic functions:

$$\overline{\overline{\xi}}_r(\mathbf{E}(\mathbf{r}),\mathbf{r},t) = \overline{\overline{\xi}}_{r_0} + \sum_{g=1}^{G}\left(\overline{\overline{\xi}}_{r_g}^{(s)}\sin(\omega_g t) + \overline{\overline{\xi}}_{r_g}^{(c)}\cos(\omega_g t)\right), \quad \overline{\overline{\xi}}_r \equiv \left\{\overline{\overline{\mu}}_r, \overline{\overline{\varepsilon}}_r\right\}$$

with

$$\overline{\overline{\xi}}_{r_0} = \frac{\omega}{2\pi}\int_0^{\frac{2\pi}{\omega}} \overline{\overline{\xi}}_r(\mathbf{E}(\mathbf{r}),\mathbf{r})\, dt$$

$$\overline{\overline{\xi}}_{r_g}^{(s)} = \frac{\omega}{\pi}\int_0^{\frac{2\pi}{\omega}} \overline{\overline{\xi}}_r(\mathbf{E}(\mathbf{r}),\mathbf{r})\sin(\omega_g t)\, dt$$

$$\overline{\overline{\xi}}_{r_g}^{(c)} = \frac{\omega}{\pi}\int_0^{\frac{2\pi}{\omega}} \overline{\overline{\xi}}_r(\mathbf{E}(\mathbf{r}),\mathbf{r})\cos(\omega_g t)\, dt$$

- S. Yamada, K. Bessho, and J. Lu, “Harmonic balance finite element method applied to nonlinear AC magnetic analysis,” *IEEE Trans. Magn.*, vol. 25, no. 4, pp. 2971–2973, Jul. 1989
- D. M. Copeland and U. Langer, “Domain decomposition solvers for nonlinear multiharmonic finite element equations,” *J. Numer. Math.*, vol. 18, no. 3, pp. 157–175, Oct. 2010

Harmonic Balance FE (multiple harmonics)

- With further Galerkin projection of the Helmholtz equation with the chosen harmonic functions ($k_0=\omega_p/c_0$), a large scale HBFE system is obtained

$$[a_{mn}][E_n]=[b_m] \quad \Rightarrow \quad a_{mn}=\begin{bmatrix} a_{mn}^{(1s1s)} & a_{mn}^{(1s1c)} & \cdots & a_{mn}^{(1sPc)} \\ a_{mn}^{(1c1s)} & a_{mn}^{(1c1c)} & \cdots & a_{mn}^{(1cPc)} \\ \vdots & \vdots & \ddots & \vdots \\ a_{mn}^{(Pc1s)} & a_{mn}^{(Pc1c)} & \cdots & a_{mn}^{(PcPc)} \end{bmatrix} \quad E_n=\begin{bmatrix} E_n^{(1s)} \\ E_n^{(1c)} \\ \vdots \\ E_n^{(Ps)} \end{bmatrix} \quad b_m=\begin{bmatrix} b_m^{(1s)} \\ b_m^{(1c)} \\ \vdots \\ b_m^{(Pc)} \end{bmatrix}$$

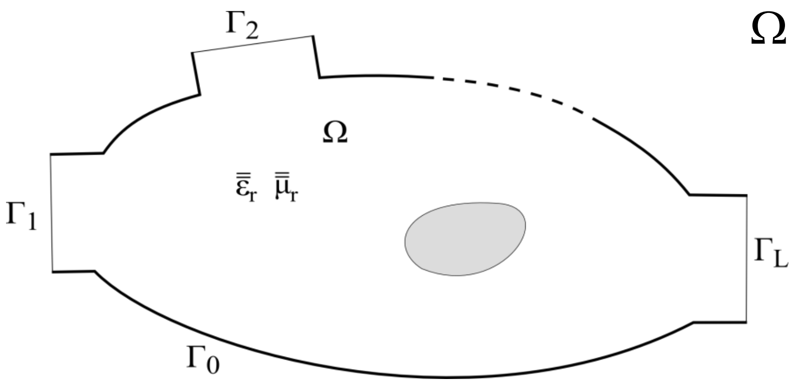
- The nonlinear solution is handled via a relaxed iteration, starting from a null solution vector to evaluate material properties

$$[A_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}][E]^i=[b_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}], \qquad i\geq 2,\ \gamma\in(0,1]$$

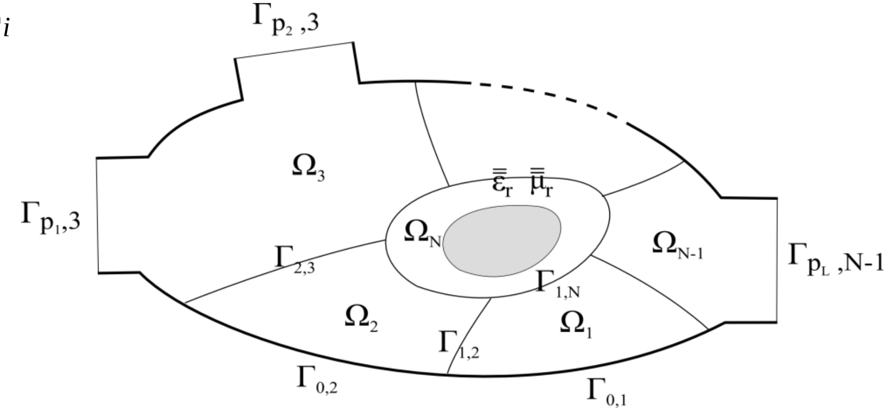
- The convergence criterion is chosen on the basis of consecutive solutions:

$$\frac{\left\|[E]^i-[E]^{i-1}\right\|}{\left\|[E]^i\right\|}\leq\tau$$

Let's assume that the device domain Ω can be decomposed into a given number of non-overlapping sub-domains Ω_i



$$\Omega = \bigcup_{i=1}^{s} \Omega_i$$



In each of these subdomains the HBFE method can be applied separately.

- G. Guarnieri, G. Pelosi, L. Rossi, S. Selleri, "A domain decomposition technique for finite element based parametric sweep and tolerance analyses of microwave passive devices," *Int. J. RF Microw. Comp. Aid. Eng.*, Vol. 19, No. 3, pp. 328-337, May 2009.

With an appropriate numbering scheme grouping all DoFs pertaining to a given interior subdomain and altogether the DoFs belonging to the domain boundaries, the following linear system of equations can be attained:

$$
\begin{bmatrix}
[M]_1 & 0 & \cdots & 0 & [P]_1 \\
0 & [M]_2 & & \vdots & [P]_2 \\
\text{[redacted]} & & & & \\
0 & \cdots & 0 & \left[M_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}\right]_N & \left[P_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}\right]_N \\
[Q]_1 & [Q]_2 & \cdots & \left[Q_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}\right]_N & \left[C_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}\right]
\end{bmatrix}
\begin{bmatrix}
[E]_1^i \\
[E]_2^i \\
\text{[redacted]} \\
[E]_N^i \\
[E]_c^i
\end{bmatrix}
=
\begin{bmatrix}
[b]_1 \\
[b]_2 \\
\vdots \\
[b_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}]_N \\
[b_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}]_c
\end{bmatrix}
$$

The system can be solved as a whole or by resorting to the Schur complement concept:

$$[S]^{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}[E]_c^i = [d]^{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}$$

It is important to notice that the Schur complement can be computed on a domain-by-domain basis as:

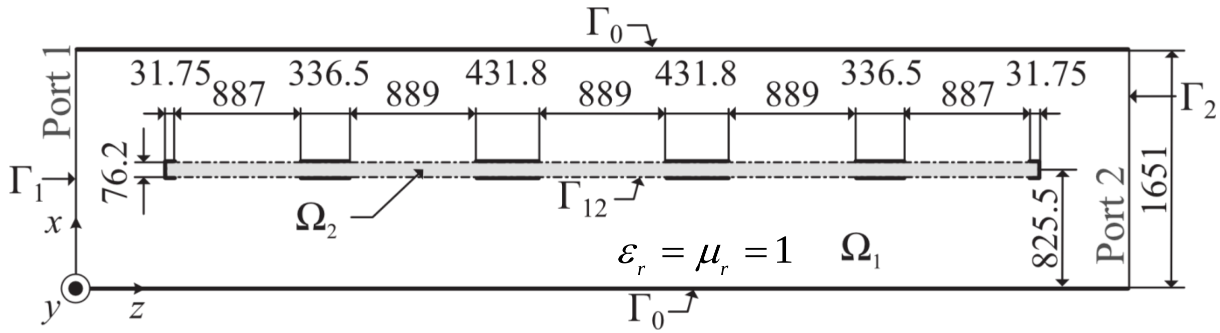
$$[S]^{i-1} = [C_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}] - \sum_{n=1}^{N-1}[Q]_n[M]_n^{-1}[P]_n - [Q_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}]_N[M_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}]_N^{-1}[P_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}]_N$$

$$[d]^{i-1} = [b_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}]_c - \sum_{n=1}^{N-1}[Q]_n[M]_n^{-1}[b]_i - [Q_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}]_N[M_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}]_N^{-1}[b_{(\gamma[E]^{i-1}+(1-\gamma)[E]^{i-2})}]_N$$

Thus, the Schur complement terms can be computed only once for all sub-domains except for the one containing the nonlinearity.

Millimeter wave filter (revisited)

- G. Guarnieri, G. Pelosi, L. Rossi, and S. Selleri, “A domain decomposition technique for efficient iterative solution of nonlinear electromagnetic problems,” *IEEE Trans. Antennas Propag.*, vol. 58, no. 12, pp. 4090–4095, Dec. 2010.



- Kerr-like permittivity of the dielectric slab:

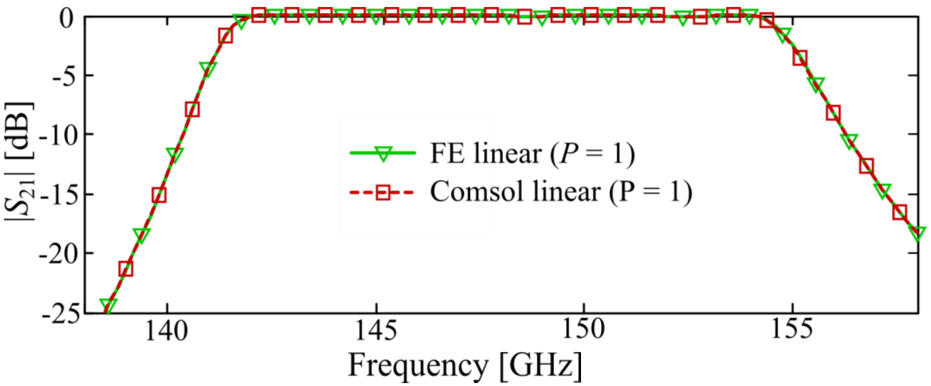
$$\overline{\overline{\varepsilon}}_r = \varepsilon_r(\mathbf{E}(\mathbf{r}),\mathbf{r}) = 2.1 + 1.625\cdot 10^{-10}\left|\mathbf{E}(\mathbf{r})\right|^2, \; \mathbf{r}\in\Omega_2$$

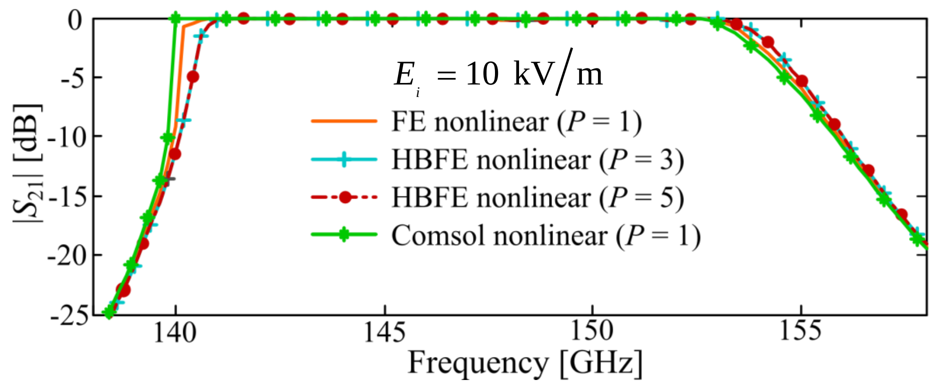
$$\mu_r(\mathbf{r}) = 1, \; \mathbf{r}\in\Omega_2$$

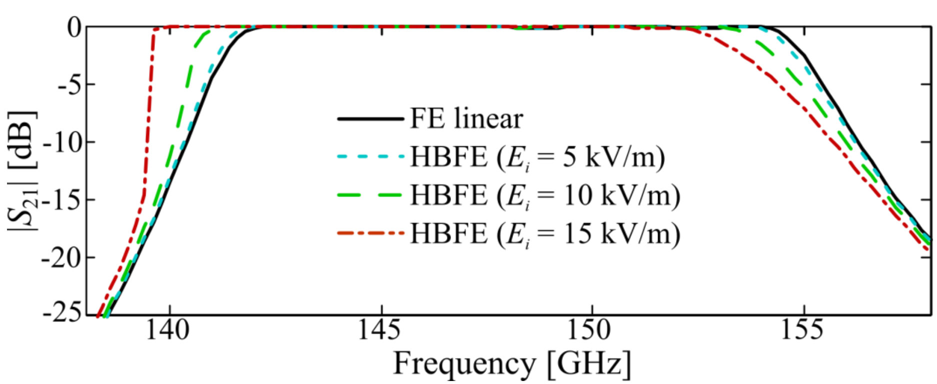
- Being the Helmholtz equation an elliptic PDE, no coupling between sine and cosine harmonic functions occurs. Either sine or cosine can be retained for field expansion of HBFE to halve matrices dimensions

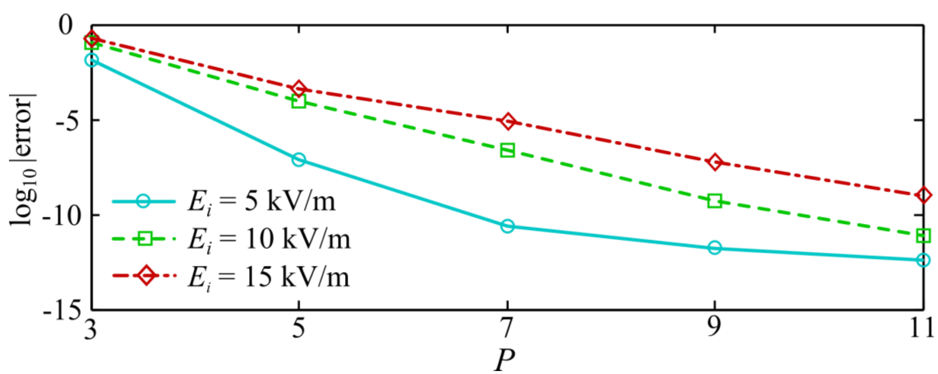
$$\hat{\mathbf{E}}(\mathbf{r},t) = \sum_{p=1}^{P} \mathbf{E}_p^{(s)}(\mathbf{r})\sin(p\omega_0 t)$$

Millimeter wave filter: linear and nonlinear frequency responses







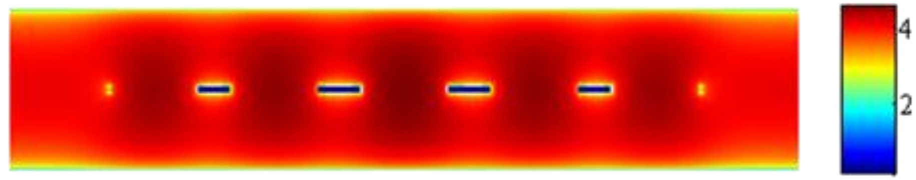


NUMERICAL EXPERIMENTS: NONLINEAR PERMITTIVITY

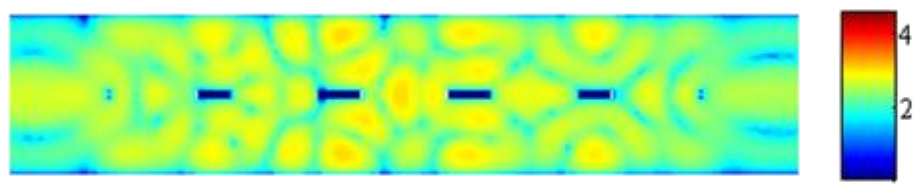
Millimeter wave filter: higher order harmonic fields

$\log_{10}|\mathbf{E}|$

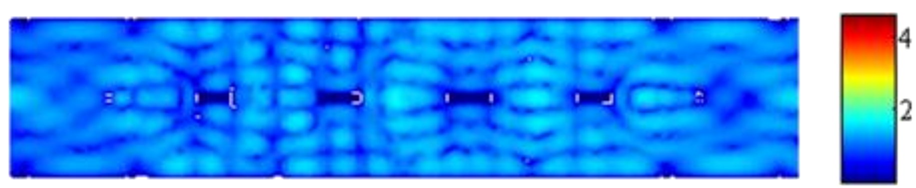
$|\mathbf{E}|$ @ 150 GHz



$|\mathbf{E}|$ @ 450 GHz



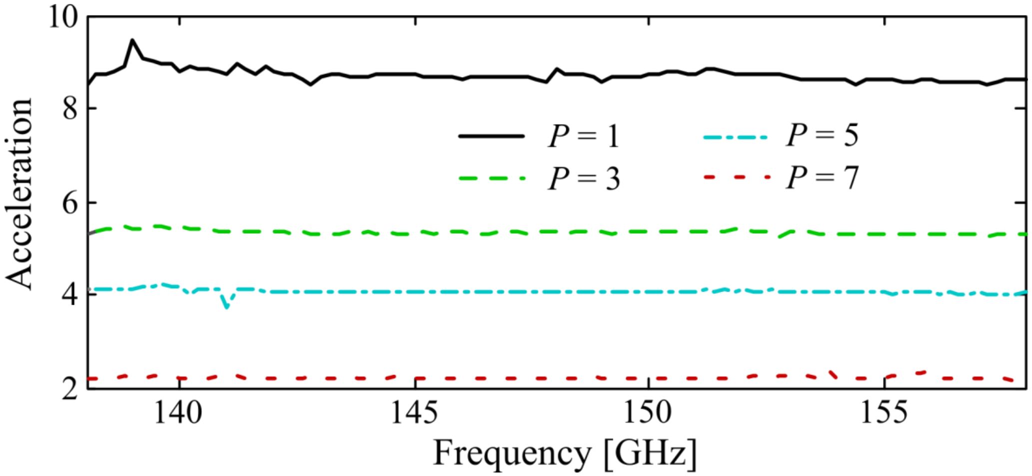
$|\mathbf{E}|$ @ 750 GHz



Millimeter wave filter: DD acceleration

Test conditions: 94 out of 2473 DoFs (conventional FE) pertain to nonlinear media ~ 3.8%

Acceleration: $\dfrac{t_{Full\ HBFE}}{t_{DD\ HBFE}}$



- L. Ntibarikure, G. Pelosi; S. Selleri, "Efficient Harmonic Balance Analysis of Waveguide Devices With Nonlinear Dielectrics," *IEEE Microw. Wireless Compon. Lett.*, vol.22, no.5, pp.221-223, May 2012

Microwave ferrite circulator: linear FE analysis

- M. Koshiba and M. Suzuki, "Finite-Element Analysis of H-Plane Waveguide Junction with Arbitrarily Shaped Ferrite Post," *IEEE Trans. Microw. Theory Tech.*, Vol. 34, no. 1, pp. 103–109, Jan. 1986.

$$\int_{\Omega} \nabla_t \varphi_m \cdot \overline{\overline{\mu}}_r^{-1} \nabla_t E_z d\Omega - k_0^2 \varepsilon_r \int_{\Omega} \varphi_m \cdot E_z d\Omega - \sum_{k=1}^{3} \int_{\Gamma_k} \varphi_m \frac{\partial E_z}{\partial n^{(k)}} d\Gamma$$

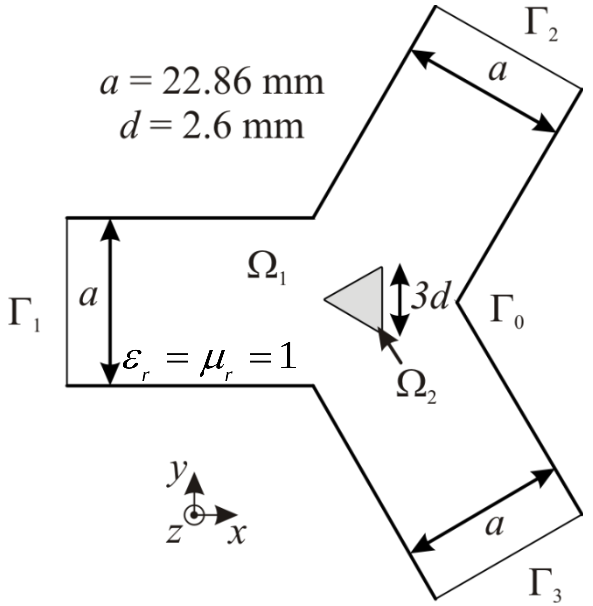
$$E_z = \sum_{n=1}^{N} E_n \varphi_n$$

$$\overline{\overline{\mu}}_r(\mathbf{r}) = \begin{bmatrix} \mu_r & j\kappa_r & 0 \\ -j\kappa_r & \mu_r & 0 \\ 0 & 0 & 1 \end{bmatrix}, \mathbf{r} \in \Omega_2$$

$$\mu_r = 1 + \frac{(\omega_0 + j\omega\alpha)\omega_m}{(\omega_0 + j\omega\alpha)^2 - \omega^2}$$

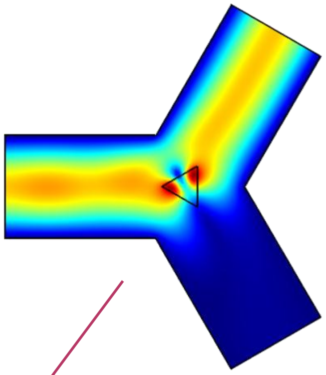
$$\kappa_r = \frac{\omega\omega_m}{(\omega_0 + j\omega\alpha)^2 - \omega^2}$$

$$\omega_0 = \gamma H_i, \quad \omega_m = \gamma M_s, \quad \alpha = \gamma \Delta H / 2\omega$$



Microwave ferrite circulator: linear FE response

The internal DC out-of-plane magnetic field H_0 is uniform



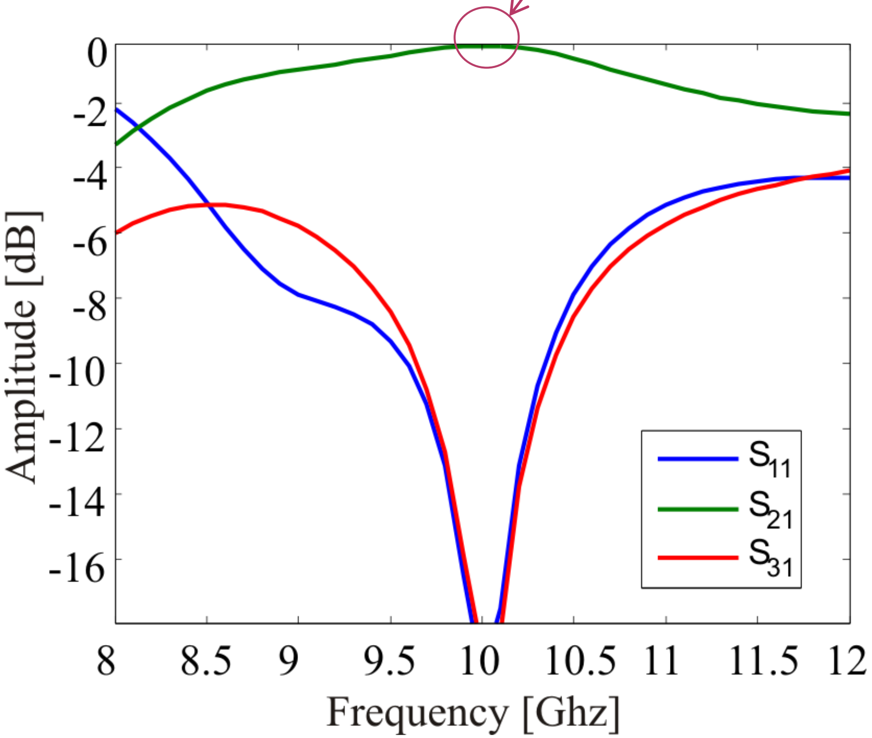
$$\gamma = 1.76 \cdot 10^{7} \left[C/kg \right]$$

$$H_i = 200 \left[G \right]$$

$$M_s = 1317 \left[\mathrm{Oe} \right]$$

$$\Delta H = 135 \left[\mathrm{Oe} \cdot \mathrm{s} \right]$$

$$\varepsilon_r \left(\mathbf{r} \right) = 11.7, \ \mathbf{r} \in \Omega_2$$



Microwave ferrite circulator: nonlinear permeability

- To derive a nonlinear permeability of the ferrite at microwave frequencies, the total magnetic field and magnetization are given by:

$$\mathbf{H} = H_i\hat{z} + \mathbf{h}_{\mathrm{AC}},\ \mathbf{M} = M_s\hat{z} + \mathbf{m}_{\mathrm{AC}}$$

- The macroscopic equation of motion is

$$\frac{\partial \mathbf{M}}{\partial t} = \mu_0\gamma\left(\mathbf{H}\times\mathbf{M}\right) \Rightarrow \begin{cases} \frac{\partial}{\partial t}m_x = \mu_0\gamma\left(h_x\left(M_s + m_z\right) - m_y\left(H_i + h_z\right)\right) \\ \frac{\partial}{\partial t}m_y = \mu_0\gamma\left(h_x\left(M_s + m_z\right) + m_x\left(H_i + h_z\right)\right) \\ \frac{\partial}{\partial t}m_z = \mu_0\gamma\left(h_x m_y - m_x h_y\right) \end{cases}$$

Microwave ferrite circulator: nonlinear permeability

- By solving the previous system of equations for each harmonic function, the following nonlinear relative permeability tensor is obtained:

$$\overline{\overline{\mu_r}}(\mathbf{H}) = \begin{bmatrix} \mu_r(\mathbf{H}) & -j\kappa_r(\mathbf{H}) & 0 \\ j\kappa_r(\mathbf{H}) & \mu_r(\mathbf{H}) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mu_r = 1 + \frac{\omega_0 \omega_m}{\omega_0^{\,2} + (\gamma h_x)^2 + (\gamma h_y)^2 - \omega^2}$$

$$\kappa_r = \frac{\omega \omega_m}{\omega_0^{\,2} + (\gamma h_x)^2 + (\gamma h_y)^2 - \omega^2}$$

- Magnetic losses are added letting

$$\omega_0 \rightarrow \omega_0 + j\omega\alpha$$

Microwave ferrite circulator: HBFE for intermodulation products

- Two signals at ω_1 and ω_2 impinging on the device are considered. The out-of-plane electric field is hence expressed in terms of harmonic functions as

$$\hat{E}_z(\mathbf{r}) = E_{z_1}^{(s)} \sin(\omega_1 t) + E_{z_2}^{(s)} \sin(\omega_2 t) + E_{z_3}^{(s)} \sin((2\omega_1 - \omega_2)t) + E_{z_4}^{(s)} \sin((2\omega_2 - \omega_1)t)$$
$$+ E_{z_1}^{(c)} \cos(\omega_1 t) + E_{z_2}^{(c)} \cos(\omega_2 t) + E_{z_3}^{(c)} \cos((2\omega_1 - \omega_2)t) + E_{z_4}^{(c)} \cos((2\omega_2 - \omega_1)t)$$

and so will be the magnetic in-plane components.

- Ferrite permeability is also expressed in terms of harmonic functions, with coefficients given by, for example for the diagonal ones of the permeability tensor:

$$\mu_{r_0} = \frac{\gcd(\omega_1, \omega_2)}{2\pi} \int_0^{\frac{2\pi}{\gcd(\omega_1, \omega_2)}} \mu_r(\mathbf{H})\, dt$$

$$\mu_{r_{m,n}^{\pm}}^{(s)} = \frac{\gcd(\omega_1, \omega_2)}{\pi} \int_0^{\frac{2\pi}{\gcd(\omega_1, \omega_2)}} \mu_r(\mathbf{H}) \sin(m\omega_1 t \pm n\omega_2 t)\, dt$$

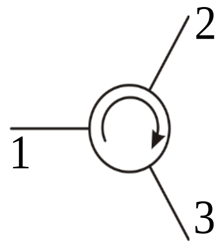
$$\mu_{r_{m,n}^{\pm}}^{(c)} = \frac{\gcd(\omega_1, \omega_2)}{\pi} \int_0^{\frac{2\pi}{\gcd(\omega_1, \omega_2)}} \mu_r(\mathbf{H}) \cos(m\omega_1 t \pm n\omega_2 t)\, dt$$

Microwave ferrite circulator: HBFE for intermodulation products

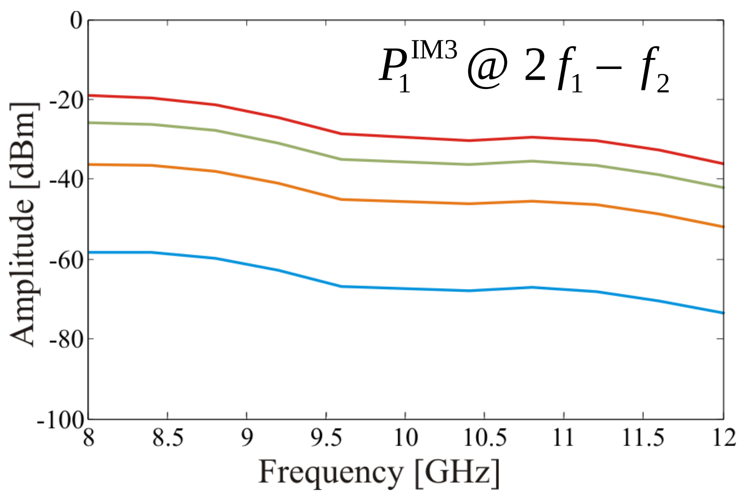
$P_1^{\mathrm{IM3}} @ \, 2f_1 - f_2$

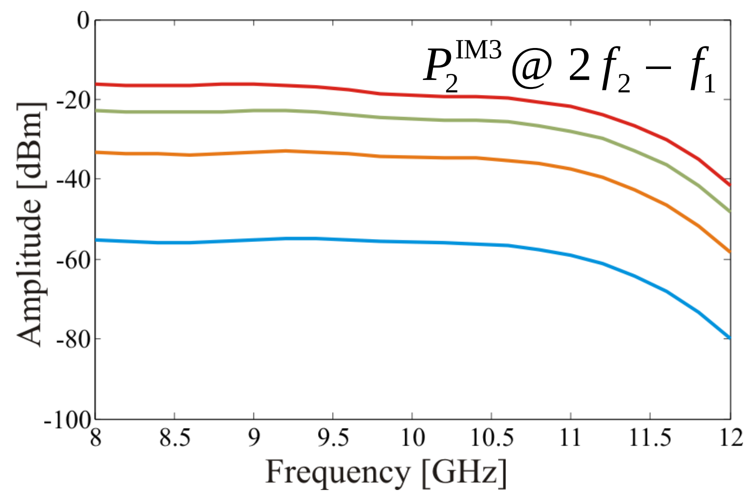
$P_2^{\mathrm{IM3}} @ \, 2f_2 - f_1$

$P_s \, @ \, f_1 \in [8,12]$ GHz



$P_i \, @ \, f_2 = 10$ GHz



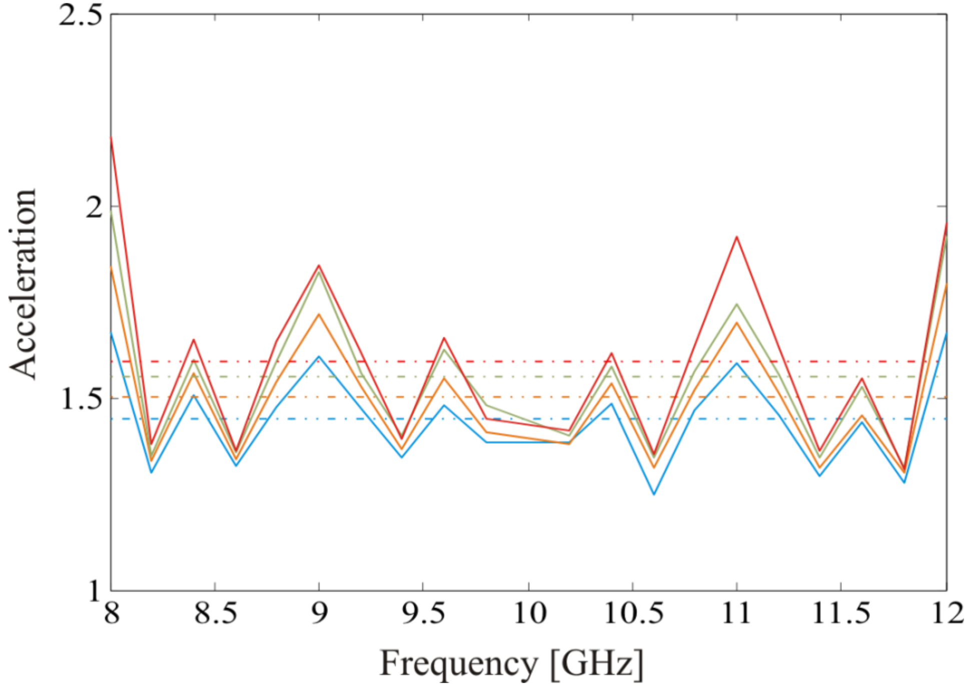


$P_s = 150$ W	$P_s = 100$ W	$P_s = 50$ W	$P_s = 10$ W
$\mathrm{P}_i = 1500$ W	$\mathrm{P}_i = 1000$ W	$\mathrm{P}_i = 500$ W	$\mathrm{P}_i = 100$ W

H. How, C. Vittoria and R. Schmidt, "Nonlinear intermodulation coupling in ferrite circulator junctions," *IEEE Trans. Microw. Theory Tech.*, Vol. 45, no. 3, pp. 245–252, Feb. 1997.

Microwave ferrite circulator: Domain decomposition

- Depending on gcd(ω_1, ω_2), the FFT algorithm employed to perform the permeability harmonic expansion required different amount of times, leading to significantly variable spectral acceleration:



- The mean acceleration has been found between 1.45 and 1.6 for the studied impinging power cases (3% of nonlinear domain)

- A Harmonic Balance Finite Element technique has been presented for solving the wave equation in presence of nonlinear media

- A substructuring domain decomposition technique have been employed to restrict the nonlinear solving loop to fewer DoFs, leading to noticeable acceleration for high nonlinearities and small nonlinear material parts